

Lab worksheet 17
Configuring & Using the DHCP service

This Lab's Objectives

1. Configure a DHCP server.
 2. Create client machines that will take their IP address from the DHCP server.
 3. Log and analyse a DHCP conversation.
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Expected Outcomes

After the laboratory session, students should be able to:

- configure a simple DHCP server to allocate IP addresses dynamically to network hosts.
 - analyse the DHCP conversation with Wireshark.
 - explain the main features of the DHCP protocol (discover, offer, request, acknowledge).
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Workplan

I recommend that you buy a hardback A4 laboratory notebook. Everything you do in the lab should be noted down for later consideration and revision. You should use your lab diary to note down items such as commands used, the outcome of each command, and answers to questions posed in these instructions.

NB this is a very complex lab – please read each instruction carefully – if you miss out any steps, then don't expect to achieve the correct results! You will need to have completed and thoroughly understood all the steps before the next lab session. Remember that there is a lab work wiki available to you on BB. Please use this to share and communicate with you colleagues.

Make sure you have completed all the steps in this worksheet before the next lab session.

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Starting assumptions

1. You have completed all the work in previous lab sheets.
 - You have used Netkit *lstart* to launch a coordinated group of several virtual machines.
 - Specifically, you are aware of how this group is coordinated via the *lab.conf* and *lab.deb* files.
 - You know how to shut down virtual machines using *lhalt* and *lcrash*.
 - You have used *ip* and *ping* within the virtual machines.
 - You have saved captured network traffic from a virtual machine using *tcpdump*.
 - You have viewed a packet capture file in the real Linux host using *wireshark* and can drill-down into the display to analyse the protocols in detail.

Activities

Reminder

- You need to make accurate notes of what you do.
- You will complete unfinished activities before the next timetabled lab session.
- You will resolve what you do not understand by conducting your own careful (and ethically sound) experimentation and / or further reading.

Getting Started

2. Create a directory for this week's work as per previous labs.
3. Download and extract the compressed file ***netkit_lab_dhcp.tar.xz*** from Blackboard.
 - This contains the configuration information for a dhcp server and two client virtual machines.
4. Personalise the comment fields in the *lab.conf* file.
5. Start the lab and check that:
 - the IP addresses are correct
 - the Netkit VMs can inter-communicate
 - the server VMs can capture packets
6. Now *lcrash* the lab to allow the configuration of the ISC DHCP server.

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Configuring the DHCP server

A DHCP server responds to requests from client machines for IP addresses, usually as the client machines boot. However, the server first needs to be configured with the range of IP addresses (the *pool*) that it can allocate.

The DHCP server available in Netkit-JH Vms is the ISC DHCP server, a widely used ,open source, service application. Read more about the server [here](#).

7. In your host terminal, move into the directory containing the netkit lab files, then create the following sub-directory:

```
.../lab_17$ mkdir -p dhcp-server/etc/dhcp
```

8. Move into the *dhcp-server/etc/dhcp/* directory, and then use the *nano* text editor to create a new file called *dhcpd.conf* and open it for editing.
NB be very careful how you spell the file name.

```
.../lab_17$ nano dhcp-server/etc/dhcp/dhcpd.conf
```

9. Add all the following content to the new file, and then save the file.

```
# dhcpd.conf
# Sample configuration file for CTEC1704 - Internet Protocols
# Netkit-JH ISC dhcpd server

ddns-update-style none;

#The server is authoritative, this DHCP server should send
#DHCPNAK messages to misconfigured clients
authoritative;

# The subnet declaration for 72.16.12.0 network
subnet 172.16.12.0 netmask 255.255.255.0
{
    range 172.16.12.65 172.16.12.150;
    option subnet-mask 255.255.255.0;
    option broadcast-address 172.16.12.255;
    option routers 172.16.12.254;

    get-lease-hostnames true;
    use-host-decl-names true;
    default-lease-time 30;
    max-lease-time 60;
}
```

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10. Spend some time understanding each of the configuration settings.
Eg *range* defines the pool of IP addresses that this DHCP server can allocate to requesting client machines.
11. What is significant about IP addresses that start with 172.16/12?
How many host addresses are available with this network prefix?
12. Review the man page for *dhcp* [here](#).

Now it is necessary to tell the DHCP server about the NIC to be used.

13. Create the following sub-directory:

```
.../lab_17$ mkdir -p dhcp-server/etc/default
```

14. Move into the *dhcp-server/etc/default/* directory, and then use the *nano* text editor to create another new file and open it for editing. The file name is *isc-dhcp-server*.
NB again, be very careful how you spell the file name.

```
.../lab_17$ nano dhcp-server/etc/default/isc-dhcp-server
```

15. Add all the following content to the new file, and then save and close the file.

```
# isc-dhcp-server
# server interface configuration

INTERFACESv4="eth0"
```

Now you are ready to test the configuration.

16. Run *lcrash/lclean* (if necessary) to halt and delete any *.disk* files, then *lstart* the lab.
17. Check that the *dhcpd.conf* file has been copied into the *dhcp-server* VM:

```
root@dhcp-server:~# cat /etc/dhcp/dhcpd.conf
```

18. In the *dhcp-server* VM start the DHCP service:

```
root@dhcp-server:~# systemctl start isc-dhcp-server.service
```

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19. Check the status of the DHCP service:

```
root@dhcp-server:~# systemctl status isc-dhcp-server.service
```

20. In the *dhcp-client-1* VM, make a note of the pre-assigned (static) IP address:

```
root@dhcp-client-1:~# ip addr show dev eth0
```

21. Remove the existing IP address:

```
root@dhcp-client-1:~# ip addr del 172.16.12.10/24 dev eth0
```

22. Then request a new IP address from the DHCP server:

```
root@dhcp-client-1:~# dhclient eth0
```

23. Recheck the IP address – it will have the first IP address from the range configured in the *dhcpd.conf* file above (172.16.12.65/24).
24. Repeat the *dhclient* command in *dhcp-client-2*, again noting the IP addresses before and after.
25. When all is working correctly, add the DHCP startup command to the *dhcp-server.startup* file:

```
systemctl start isc-dhcp-server
```

Reset the networking environment

The two client VMs currently have static IP addresses, configured in the relevant startup file. Their configurations need to be changed to automatically request an IP address from the new DHCP server as they start up.

26. Crash the lab.
27. Open the two client *.startup* files in an editor.
28. In each file, comment out the static IP address configuration line (with the hash “#” character).

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29. Add the line:

```
dhclient eth0
```

to each client startup file.

30. Start ONLY the *dhcp-server* VM:

```
.../lab_17$ lstart dhcp-server
```

31. Start a *tcpdump* capture in the *dhcp-server* VM.

```
dhcp-server:~# tcpdump -i eth0 -s0 -v -w  
/hosthome/ctec1907/captures/dhcp1.pcap
```

32. Restart the client machines one at a time.

In each case, capture the DHCP conversation for analysis, with *tcpdump*. Leave *tcpdump* running for a few minutes at least.

```
.../lab_17$ lstart dhcp-client-1  
...  
.../lab_17$ lstart dhcp-client-2
```

33. Use *wireshark* on the host system to analyse the pcap file.
Look for the DORA cycle packets – drill down into the header field values for each to see the values as discussed in the lecture.

Set the filter in Wireshark to be

```
bootp.option.type == 53
```

The display will change to show DHCP packets only (why?).

34. If you left the *tcpdump* capture running for long enough, you will also see the re-negotiation cycle for the IP address taking place as the lease expires. Which DORA messages are used to achieve this?
35. Restart the lab. Look out for any error messages ...
36. One potential problem is that the DHCP server has to start first, so that it is ready to respond to requests from client machines. Read the *lab.dep* to see how this is achieved.

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Monitor the log files

As machines request addresses from the DHCP server, they will leave a trace in the `/var/log/daemon.log` file. NB this is specific to the Linux used by netkit - in other Linux distributions, this access may be logged in another system log file.

37. Monitor this file (in the *dhcp-server* VM) to see what data is written to the log when a client requests an IP address. Read the *man tail* page and make notes.

```
dhcp-server:~# tail -f /var/log/daemon.log
```

Give the client virtual machines a MAC address

Netkit automatically assigns each virtual machine with a MAC address, but you are going to manually set a MAC address for the client machines. For each client machine you will need to edit the *startup* file: insert the hardware address command as the first line.

```
ip link set dev eth0 address XX:XX:XX:XX:XX:XX
ip link set dev eth0 up
dhclient eth0
```

For each client machines, replace the "xx:xx..." with the respective MAC address as below:

```
dhcp-client-1: 02:0f:c9:02:02:02
dhcp-client-2: 02:0f:c9:04:04:04
```

38. Restart each client machines, capturing each packet flow again. Ensure that you can see the new MAC addresses in the pcap files.
39. Create an extra client VM (*dhcp-client-3*), with the MAC address:
- ```
02:0f:c9:06:06:06
```
40. Don't forget to include this new machine in the *lab.conf* and *lab.dep* files.
41. Use *lcrash* and *lstart* to include the new machine.

**Challenges:**

42. Create a second dhcp server on the same subnet. Use the same *dhcpd.conf* file settings, but give each server a different range of pool IP addresses.
43. Capture all messages as before, in each of the two servers. Analyse both capture files – what differences do you note?
44. Which server leased IP addresses to your clients?

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45. Research how to set a DHCP static lease for a particular MAC address and configure the lab so that *dhcp-client-1* and *dhcp-client-2* will be given their original IP addresses as originally configured when they request an IP address from a DHCP server (solution in a later lab).
46. There is a configuration file for the client machine which allows the client to specify exactly what optional information it wishes to receive. The configuration file is at *etc/dhcp/dhclient.conf*. There is a manual page for this configuration [here](#). If you have time, experiment!
47. Now review what you have learned in this lab session.

**Make sure you have completed all the steps in this worksheet before the next lab session.**